



EKAGRA

Threat detection that survives a one-way tap

Deeper detail: last line of page 2

The problem

Defence, power-grid and classified networks are watched through a **data diode**: hardware that copies traffic out to a monitoring room and allows nothing back. The sensor there never sees replies, cannot look anything up, and cannot block. Detectors are trained on two-way traffic, so they fail there - and with no return path, nobody can tell them they are wrong.

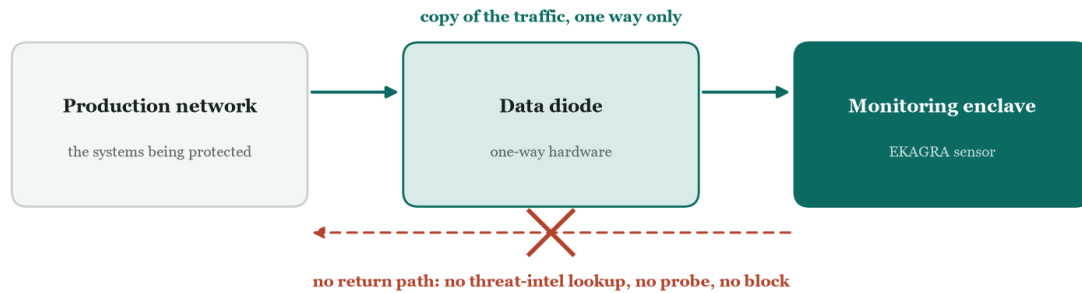
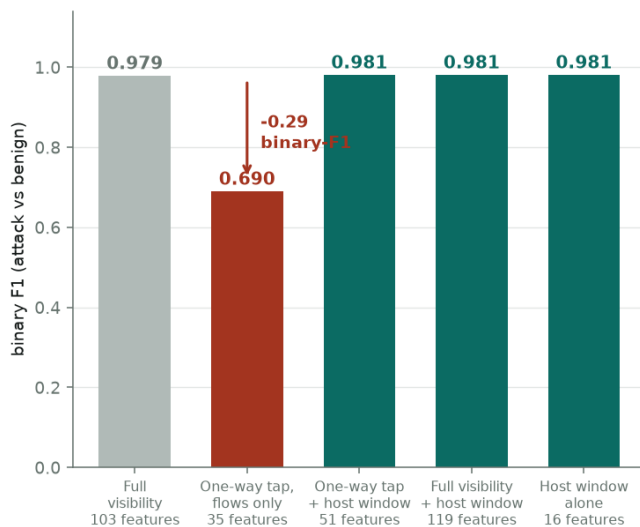


Figure 1. The deployment NTRO's problem statement describes: a copy of the traffic goes out, nothing comes back.

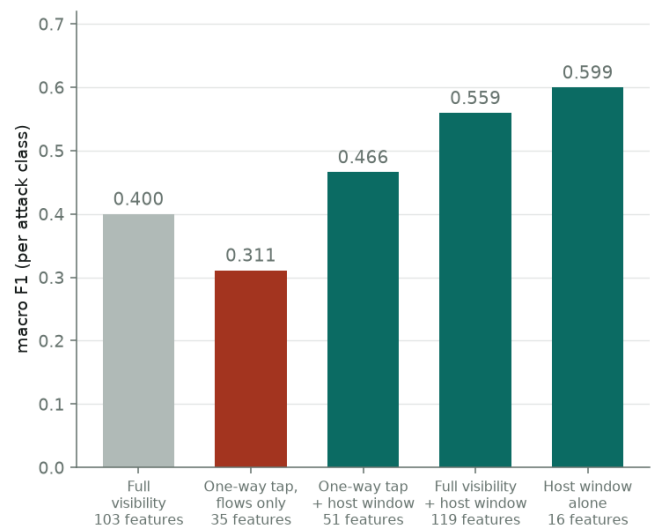
What we measured, on 877,801 real flows

CIC-IDS2017: what a one-way tap actually costs

Losing the reverse direction costs nothing — provided the sensor keeps host state



On multi-class the host window is not a patch, it is the signal



877,801 flows, 60s windows. 16 host-window features alone (0.599 macro-F1) beat all 103 full-visibility features (0.400).

Figure 2. CIC-IDS2017 (corrected re-extraction), 877,801 flows, temporal split, IP addresses excluded. Red: a conventional detector behind the tap.

The loss is not the missing direction. It is missing **per-host memory**. A sensor that keeps 60 seconds of behaviour per host - peers touched, timing regularity, volume against its own baseline - recovers all of it.

0.690

conventional detector
on a one-way tap

0.981

EKAGRA on the
same traffic

9.1%

cases where it says
"I don't know"

17 of 42

own predictions refuted
and published

What is built, and proved

Capability	Evidence	Measured	Value
Six threat detectors	the six classes the PS names	Throughput	610 to 672 Mbps (not the 1 Gbps we once claimed)
Severity + calibrated confidence	kept separate, as CVSS does	Alert latency	at most 75 s
Abstains when unsure	9.11% of cases, 99.55% right on the rest	Sensor memory	2.66 MB for 14,817 hosts
Tamper-evident evidence	SHA-256 chain + HMAC, 7 attack tests	Calibration	coverage 0.908, ECE 0.00084
Verified in the browser	console re-hashes independently, offline	Tests passing	290
Cannot phone home	a test fails the build on a network import	Hardware	one commodity server, no GPU

Why you can trust these numbers

Every experiment's predictions were written down *before* it ran, and the run prints the verdicts. Seventeen of forty-two failed, including the claim this project started with:

What we predicted	What happened
Our custom one-way feature set beats simple retraining	It did not. A control showed host memory alone did the work, so we rewrote the claim
Eight shards reach 1 Gbps	Not reached. 108,938 packets/s on twelve shards
A threshold tuned on generated benign domains transfers to real ones	It flagged 89.6% of real benign names, so every threshold is now set on real traffic

Who buys it, and what it costs

Buyer	Why	Five years, one enclave	Cost
NTRO and NCIIPC	the problem owner; eight critical sectors	EKAGRA	about Rs 9.5 lakh (server + operator time)
Power, rail, telecom	control networks already behind diodes	Commercial NDR	about Rs 132 lakh at list price
Defence and classified	monitoring must never become a way in	Licence fee	zero, and no data leaves the enclave

Cost assumes 2,000 IPs at \$15 per IP per year, Rs 88 to the dollar, no discount. CERT-In handled 29.44 lakh incidents in 2025, up 85% on 2023, and its 2022 directions require 180 days of securely held logs - which is what the evidence chain is for.

What is missing, and what is next

HONEST LIMITS Packet-size and timing sequences are not built yet, and the problem statement names them. Dictionary-based DGA domains are weak (AUC 0.69). Evidence integrity depends on a key held in the enclave. 1 Gbps is not reached.	NEXT, IN ORDER 1. Packet-size and timing features. 2. A compiled fast path to line rate. 3. Longer real captures for DNS thresholds. 4. Hardware key custody. 5. Pilot on vendor diode hardware.
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Going deeper: the full 66-page report (method, threat model, risk register, compliance) is [EKAGRA_Comprehensive_Report_SIH26145.pdf](#); code and every pre-registered verdict are at [github.com/AKRai-2005/ekagra](#). The same report is also readable as a web page.